Multivariate Probability Distributions

Independent Random Variables, Expected Value

Independent Random Variables

Introduction

Two events A and B are independent

$$\text{if } P(A \cap B) = P(A) \times P(B).$$

When discussing random variables,

if $a < b$ and $c < d$ we are often concerned with events of the type

$$(a < Y_1 \le b) \cap (c < Y_2 \le d).$$

For consistency with the earlier definition of independent events, if Y1 and Y2 are independent, we would like to have

$$P(a < Y_1 \leq b, c < Y_2 \leq d) = P(a < Y_1 \leq b) \times P(c < Y_2 \leq d)$$

for any choice of real numbers $a < b$ and $c < d$.

That is, if Y_1 and Y_2 are independent, the joint probability can be written as the product of the marginal probabilities.

DEFINITION 8

Let Y_1 have distribution function $F_1(y_1)$, Y_2 have distribution function $F_2(y_2)$, and Y_1 and Y_2 have joint distribution function $F(y_1, y_2)$. Then Y_1 and Y_2 are said to be independent if and only if

$$F(y_1, y_2) = F_1(y_1)F_2(y_2)$$

for every pair of real numbers (y_1, y_2). If Y_1 and Y_2 are not independent, they are said to be dependent.

THEOREM 4

If Y_1 and Y_2 are discrete random variables with joint probability function $p(y_1, y_2)$ and marginal probability functions $p_1(y_1)$ and $p_2(y_2)$, respectively, then Y_1 and Y_2 are independent if and only if

$$p(y_1, y_2) = p_1(y_1)p_2(y_2)$$

for all pairs of real numbers (y_1, y_2).

If Y_1 and Y_2 are continuous random variables with joint density function $f(y_1, y_2)$ and marginal density functions $f_1(y_1)$ and $f_2(y_2)$, respectively, then Y_1 and Y_2 are independent if and only if

$$f(y_1, y_2) = f_1(y_1)\, f_2(y_2)$$

for all pairs of real numbers (y_1, y_2).

Example 8

For the die-tossing problem of previous section , show that Y1 and Y2 are independent.

Example 9

Refer to Example 4, Is the number of Republicans in the sample independent of the number of Democrats? (Is Y_1 independent of Y_2 ?)

Example 10

Let

$$f(y_1, y_2) = \begin{cases} 6y_1 y_2^2, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that Y_1 and Y_2 are independent.

Example 11

Let

$$f(y_1, y_2) = \begin{cases} 2, & 0 \le y_2 \le y_1 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that Y_1 and Y_2 are dependent.

THEOREM 5

Let Y_1 and Y_2 have a joint density $f(y_1, y_2)$ that is positive if and only if

$$a \le y_1 \le b \text{ and } c \le y_2 \le d,$$

for constants a, b, c, and d; and $f(y_1, y_2) = 0$ otherwise.

Then Y_1 and Y_2 are independent random variables if and only if

$$f(y_1, y_2) = g(y_1)h(y_2)$$

where $g(y_1)$ is a nonnegative function of y_1 alone and $h(y_2)$ is a nonnegative function of y_2 alone.

Example 12

Let Y_1 and Y_2 have a joint density given by

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Are Y_1 and Y_2 independent variables?

Definition 8 easily can be generalized to n dimensions.

Suppose that we have n random variables, $Y_1, \ldots, Y_n$, where Y_i has distribution function $F_i(y_i)$, for $i = 1, 2, \ldots, n$; and where $Y_1, Y_2, \ldots, Y_n$ have joint distribution function $F(y_1, y_2, \ldots, y_n)$.

Then $Y_1, Y_2, \ldots, Y_n$ are independent if and only if

$$F(y_1, y_2, \ldots, y_n) = F_1(y_1) \cdots F_n(y_n)$$

for all real numbers $y_1, y_2, \ldots, y_n$, with the obvious equivalent forms for the discrete and continuous cases.

Q1

Let Y_1 and Y_2 have joint density function $f(y_1, y_2)$ and marginal densities $f_1(y_1)$ and $f_2(y_2)$, respectively. Show that Y_1 and Y_2 are independent if and only if $f(y_1|y_2) = f_1(y_1)$ for all values of y_1 and for all y_2 such that $f_2(y_2) > 0$.

A completely analogous argument establishes that Y_1 and Y_2 are independent if and only if $f(y_2|y_1) = f_2(y_2)$ for all values of y_2 and for all y_1 such that $f_1(y_1) > 0$.

Q2

Suppose that the probability that a head appears when a coin is tossed is p and the probability that a tail occurs is $q = 1 - p$. Person A tosses the coin until the first head appears and stops. Person B does likewise. The results obtained by persons A and B are assumed to be independent. What is the probability that A and B stop on exactly the same number toss?

Group Project (Ex)

A supermarket has two customers waiting to pay for their purchases at counter I and one customer waiting to pay at counter II. Let Y_1 and Y_2 denote the numbers of customers who spend more than \$50 on groceries at the respective counters. Suppose that Y_1 and Y_2 are independent binomial random variables, with the probability that a customer at counter I will spend more than \$50 equal to .2 and the probability that a customer at counter II will spend more than \$50 equal to .3. Find the

1) joint probability distribution for Y_1 and Y_2 .
2) probability that not more than one of the three customers will spend more than \$50.

Expected Value of function of random variables

DEFINITION 9

Let $g(Y_1, Y_2, \ldots, Y_k)$ be a function of the discrete random variables, $Y_1, Y_2, \ldots, Y_k$, which have probability function $p(y_1, y_2, \ldots, y_k)$. Then the expected value of $g(Y_1, Y_2, \ldots, Y_k)$ is

$$E[g(Y_1, Y_2, \ldots, Y_k)] = \sum_{\text{all } y_k} \cdots \sum_{\text{all } y_2} \sum_{\text{all } y_1} g(y_1, y_2, \ldots, y_k) p(y_1, y_2, \ldots, y_k).$$

If Y1, Y2 ,...., Yk are continuous random variables with joint density function f (y1, y2 ,...., yk), then

$$E[g(Y_1, Y_2, \ldots, Y_k)] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(y_1, y_2, \ldots, y_k) \times f(y_1, y_2, \ldots, y_k)\, dy_1\, dy_2 \ldots dy_k.$$

Example 13

Let Y_1 and Y_2 have a joint density given by

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $E(Y_1Y_2)$.

Consider two random variables Y_1 and Y_2 with density function f (y_1, y_2).
We wish to find the expected value of $g(Y_1, Y_2) = Y_1$.
Then from Definition 9 we have

$$E(Y_1) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y_1 f(y_1, y_2)\, dy_2\, dy_1$$

$$= \int_{-\infty}^{\infty} y_1 \left[\int_{-\infty}^{\infty} f(y_1, y_2)\, dy_2\right] dy_1.$$

The quantity within the brackets, by definition, is the marginal density function for Y_1. Therefore, we obtain

$$E(Y_1) = \int_{-\infty}^{\infty} y_1 f_1(y_1)\, dy_1,$$

Example 14

Let Y_1 and Y_2 have a joint density given by

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of Y_1

Example 15

Let Y_1 and Y_2 have a joint density given by

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $V(Y_1)$.

Example 16

A process for producing an industrial chemical yields a product containing two types of impurities. For a specified sample from this process, let Y_1 denote the proportion of impurities in the sample and let Y_2 denote the proportion of type I impurities among all impurities found. Suppose that the joint distribution of Y_1 and Y_2 can be modeled by the following probability density function:

$$f(y_1, y_2) = \begin{cases} 2(1 - y_1), & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of the proportion of type I impurities in the sample.

Special Theorems

THEOREM 6

Let c be a constant. Then

$$E(c) = c.$$

THEOREM 7

Let $g(Y_1, Y_2)$ be a function of the random variables Y_1 and Y_2 and let c be a constant. Then

$$E[cg(Y_1, Y_2)] = cE[g(Y_1, Y_2)].$$

THEOREM 8

Let Y_1 and Y_2 be random variables and

$g_1(Y_1, Y_2), g_2(Y_1, Y_2), \ldots, g_k(Y_1, Y_2)$ be functions of Y_1 and Y_2. Then

$$E[g_1(Y_1, Y_2) + g_2(Y_1, Y_2) + \cdots + g_k(Y_1, Y_2)] = E[g_1(Y_1, Y_2)] + E[g_2(Y_1, Y_2)] + \cdots + E[g_k(Y_1, Y_2)].$$

Example 17

Suppose that the joint distribution of Y_1 and Y_2 can be modeled by the following probability density function:

$$f(y_1, y_2) = \begin{cases} 3y_1, & 0 \le y_2 \le y_1 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Consider the random variable $Y_1 - Y_2$. Find $E(Y_1 - Y_2)$.

THEOREM 9

Let Y_1 and Y_2 be independent random variables and $g(Y_1)$ and $h(Y_2)$ be functions of only Y_1 and Y_2 , respectively. Then

$$E[g(Y_1)h(Y_2)] = E[g(Y_1)]E[h(Y_2)],$$

provided that the expectations exist.

Proof

We will give the proof of the result for the continuous case. Let f (y_1, y_2) denote the joint density of Y_1 and Y_2 . The product $g(Y_1)h(Y_2)$ is a function of Y_1 and Y_2 . Hence, by Definition 9 and the assumption that Y_1 and Y_2 are independent,

$$
\begin{aligned}
E[g(Y_1)h(Y_2)] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(y_1)h(y_2)f(y_1, y_2)\,dy_2\,dy_1 \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(y_1)h(y_2)f_1(y_1)f_2(y_2)\,dy_2\,dy_1 \\
&= \int_{-\infty}^{\infty} g(y_1)f_1(y_1)\left[\int_{-\infty}^{\infty} h(y_2)f_2(y_2)\,dy_2\right]dy_1 \\
&= \int_{-\infty}^{\infty} g(y_1)f_1(y_1)E[h(Y_2)]\,dy_1 \\
&= E[h(Y_2)]\int_{-\infty}^{\infty} g(y_1)f_1(y_1)\,dy_1 = E[g(Y_1)]\,E[h(Y_2)].
\end{aligned}
$$

Example 18

Suppose that Y_1 and Y_2 are independent random variables. Find $E(Y_1Y_2)$ by using the result that $E(Y_1Y_2) = E(Y_1)E(Y_2)$ if Y_1 and Y_2 are independent.

Q1

In Q1 in 1.1, we determined that the joint distribution of Y_1, the number of contracts awarded to firm A, and Y_2 , the number of contracts awarded to firm B, is given by the entries in the following table.

	y_1		
y_2	0	1	2
0	1/9	2/9	1/9
1	2/9	2/9	0
2	1/9	0	0

The marginal probability function of Y_1 was derived in previous Exercise to be binomial with n = 2 and p = 1/3.

Find

a $E(Y_1)$.

b $V(Y_1)$.

c $E(Y_1 - Y_2)$.

Q2

we derived the fact that

$$f(y_1, y_2) = \begin{cases} 4y_1y_2, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

a Find $E(Y_1)$.
b Find $V(Y_1)$.
c Find $E(Y_1 - Y_2)$.